IP Address & Subnetting Questions

9 questions | Original English questions and source answers

Question 1

Refer to exhibit. Which statement explains the configuration error message that is received?

```
Router(config)#interface GigabitEthernet 1/0/1
Router(config-if)#ip address 192.168.16.143 255.255.255.240
Bad mask /28 for address 192.168.16.143
```

Options

A. It belongs to a private IP address range

B. It is a network IP address

C. It is a broadcast IP address

D. The router does not support /28 mask

Answer

C. It is a broadcast IP address

Source: CCNA 200-301 Questions.html | Question 1

Question 2

Which network allows devices to communicate without the need to access the Internet?

Options

A. 192.0.0.0/8

B. 209.165.201.0/24

C. 172.28.0.0/16

D. 172.9.0.0/16

Answer

C. 172.28.0.0/16

Source explanation

This question asks about the private ranges of IPv4 addresses. The private ranges of each class of IPv4 addresses:

Class A private IP address ranges from 10.0.0.0 to 10.255.255.255
Class B private IP address ranges from 172.16.0.0 to 172.31.255.255
Class C private IP address ranges from 192.168.0.0 to 192.168.255.255

Only the network 172.28.0.0/16 belongs to the private IP address (of class B).

Source: CCNA 200-301 Questions.html | Question 2

Question 3

An office has 8 floors with approximately 30-40 users per floor. What command must be configured on the router Switched Virtual Interface to use address space efficiently?

Options

A. ip address 192.168.0.0 255.255.254.0

B. ip address 192.168.0.0 255.255.255.128

C. ip address 192.168.0.0 255.255.255.224

D. ip address 192.168.0.0 255.255.0.0

Answer

C. ip address 192.168.0.0 255.255.255.224

Source explanation

We cannot use the subnet mask of 255.255.254.0 (/23) because it is smaller than the default subnet mask of class C (/24). If we use /23 subnet mask, in fact we are summarizing (supernet) it -> Answer "ip address 192.168.0.0 255.255.254.0" is not correct.

With the subnet mask of 255.255.255.128, we have 2^1 = 2 subnets only, not enough for 8 floors -> Answer "ip address 192.168.0.0 255.255.255.128" is not correct.

We need 40 users per subnet so our subnet must support at least 64 (=2^6) hosts (in fact 62 hosts). So the last octet of subnet mask should be 1100 0000 (with 6 bits “0”) = 192 -> The suitable subnet mask is 255.255.255.192. But this subnet mask only provides 2^2 = 4 subnets, which is not enough for 8 floors.

For 8 floors we need 1110 0000 (2^3 = 8) for the last octet of subnet mask so the subnet mask should be 255.255.255.224.

But with the subnet mask of 255.255.255.224, the host addresses per subnet is 2^5 - 2 = 30 so we cannot fulfill the host addresses for 40 users. We can only provide IPs for 30 users. This question said "approximately 30-40 users per floor" so "30 users per floor" is acceptable.

With “192.168.0.0 255.255.255.224” we have 8 subnets:
+ First subnet: 192.168.0.0 to 192.168.0.31
+ Second subnet: 192.168.0.32 to 192.168.0.63
+ Third subnet: 192.168.0.64 to 192.168.0.95
+ Fourth subnet: 192.168.0.96 to 192.168.0.127
+ Fifth subnet: 192.168.0.128 to 192.168.0.159
+ Sixth subnet: 192.168.0.160 to 192.168.0.191
+ Seventh subnet: 192.168.0.192 to 192.168.0.223
+ Eighth subnet: 192.168.0.224 to 192.168.0.255

Source: CCNA 200-301 Questions.html | Question 3

Question 4

Which function does the range of private IPv4 addresses perform?

Options

A. provides a direct connection for hosts from outside of the enterprise network

B. allows multiple companies to each use the same addresses without conflicts

C. ensures that NAT is not required to reach the internet with private range addressing

D. enables secure communications to the internet for all external hosts

Answer

B. allows multiple companies to each use the same addresses without conflicts

Source: CCNA 200-301 Questions.html | Question 4

Question 5

A corporate office uses four floors in a building
* Floor 1 has 24 users
* Floor 2 has 29 users
* Floor 3 has 28 users
* Floor 4 has 22 users
Which subnet summarizes and gives the most efficient distribution of IP addresses for the router configuration?

Options

A. 192.168.0.0/23 as summary and 192.168.0.0/25 for each floor

B. 192.168.0.0/25 as summary and 192.168.0.0/27 for each floor

C. 192.168.0.0/26 as summary and 192.168.0.0/29 for each floor

D. 192.168.0.0/24 as summary and 192.168.0.0/28 for each floor

Answer

B. 192.168.0.0/25 as summary and 192.168.0.0/27 for each floor

Source explanation

We see the maximum number of user per floor is 29 users (Floor 2) < 32 so the best subnet mask should be 1110 0000 which allows 2^5 – 2 = 30 hosts per subnet.

Source: CCNA 200-301 Questions.html | Question 5

Question 6

Which goal is achieved by the implementation of private IPv4 addressing on a network?

Options

A. provides a reduction in size of the forwarding table on network routers

B. provides an added level of protection against Internet exposure

C. allows servers and workstations to communicate across public network boundaries

D. allows communication across the Internet to other private networks

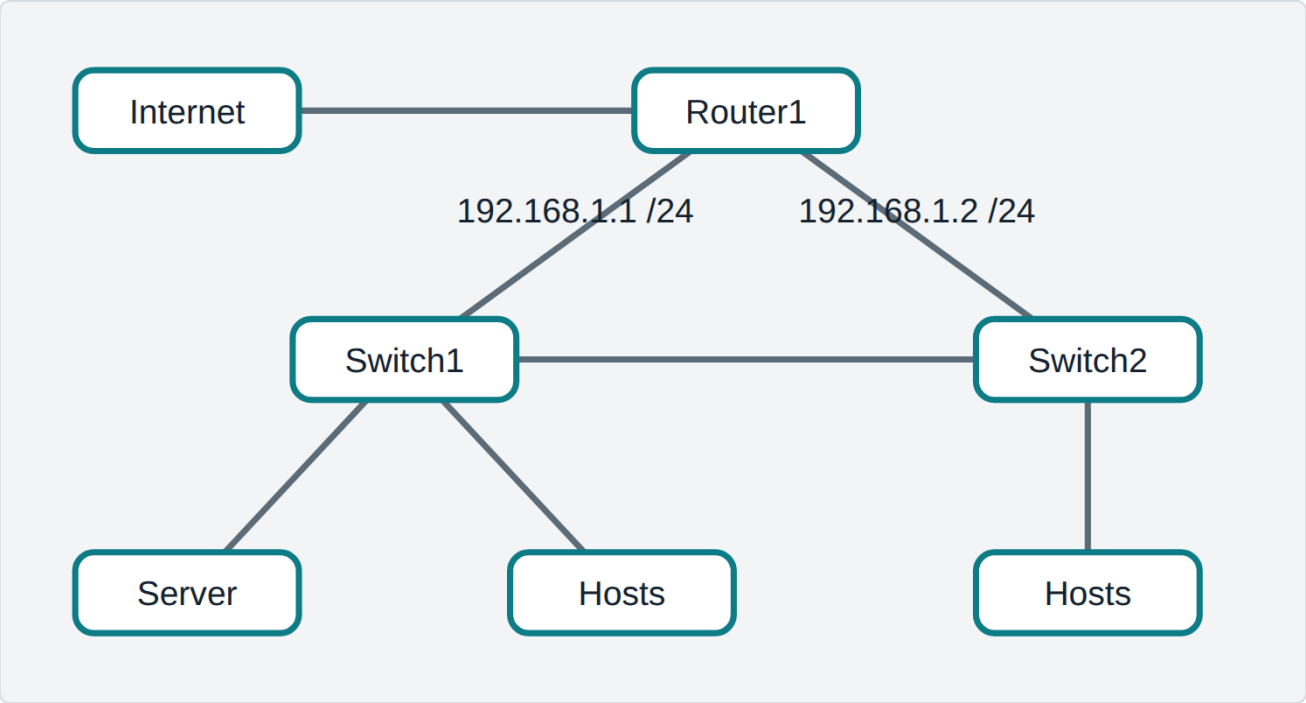
Answer

B. provides an added level of protection against Internet exposure

Source: CCNA 200-301 Questions.html | Question 6

Question 7

Refer to the exhibit. A network technician is asked to design a small network with redundancy. The exhibit represents this design, with all hosts configured in the same VLAN. What conclusions can be made about this design?



Options

A. The design will function as intended

B. The connection between switches should be a trunk.

C. The router interfaces must be encapsulated with the 802.1Q protocol.

D. Spanning-tree will need to be used.

E. The router will not accept the addressing scheme.

Answer

E. The router will not accept the addressing scheme.

Source explanation

Each interface on a router must be in a different network. If two interfaces are in the same network, the router will not accept it and show error when the administrator assigns it.

Source: CCNA 200-301 Questions.html | Question 7

Question 8

An engineer must configure a /30 subnet between two routers. Which usable IP address and subnet mask combination meets this criteria?

Options

```
A. interface e0/0
description to HQ-A371:10975
ip address 10.2.1.3 255.255.255.252
```

```
B. interface e0/0
description to HQ-A371:10975
ip address 172.16.1.4 255.255.255.248
```

```
C. interface e0/0
description to HQ-A371:10975
ip address 192.168.1.1 255.255.255.248
```

```
D. interface e0/0
description to HQ-A371:10975
ip address 209.165.201.2 255.255.255.252
```

Answer

D. interface e0/0
description to HQ-A371:10975
ip address 209.165.201.2 255.255.255.252

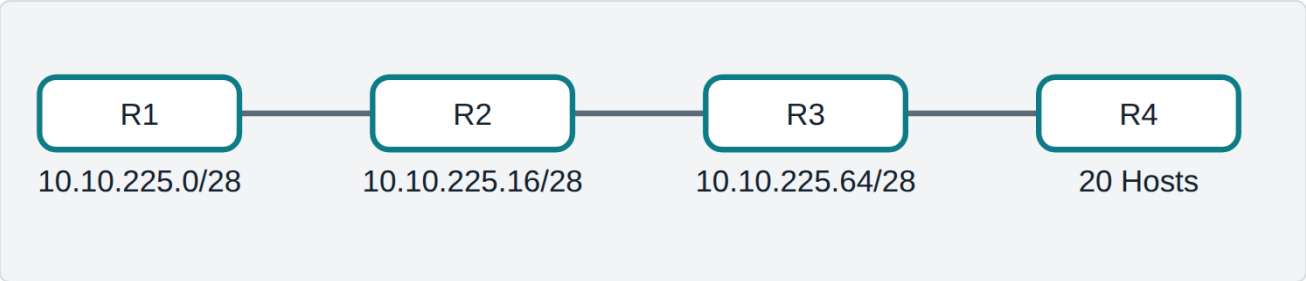
Source explanation

A /30 subnet means subnet mask of 255.255.255.252. But 10.2.1.3 255.255.255.252 is a broadcast IP address; only 209.165.201.2/30 is the usable IP address.

Source: CCNA 200-301 Questions.html | Question 8

Question 9

Refer to the exhibit. An engineer must add a subnet for a new office that will add 20 users to the network. Which IPv4 network and subnet mask combination does the engineer assign to minimize wasting addresses?



Options

A. 10.10.225.48 255.255.255.224

B. 10.10.225.32 255.255.255.224

C. 10.10.225.48 255.255.255.240

D. 10.10.225.32 255.255.255.240

Answer

B. 10.10.225.32 255.255.255.224

Source explanation

We need a subnet with 20 users so we need 5 bits 0 in the subnet mask as 2^5 – 2 = 30 > 20. Therefore the subnet mask should be /27 (with last octet is 1110 0000 in binary). The increment is 32 so the valid network address is 10.10.225.32.

Source: CCNA 200-301 Questions.html | Question 9